

Notes

What you already solved

You have essentially done the first component:

Generate distinct unions/signatures efficiently

I implemented the distinct-union generation component using the naive method and the flashlight/backtracking approach, with an extension test and benchmark.

So now what are we actually building?

The input is a binary polynomial f and a subset X of variables that we want to eliminate. The goal is to compute the projection onto the remaining variables Y , where for every fixed Y -assignment we optimize over the X -variables.

Here by “eliminate,” we mean:

We don't just delete the X -variables. We maximize over them, so their optimal contribution is preserved in the projected polynomial.

One very reasonable first design is:

```
INPUT
  polynomial  $f$ 
  subset  $X$ 
    ↓
  Parse monomials as
  (coefficient, variable-set)
    ↓
  Compute  $Y = V \setminus X$ 
    ↓
  Build  $P, Q, R$ 
    ↓
  Build  $T$  from  $Q$ 
    ↓
  Generate distinct signatures of  $T$ 
```

```

↓
For each signature y:
  construct βy
  compute μ(y)
↓
Compute δ coefficients
↓
Construct output monomials
↓
Combine equal monomials
↓
OUTPUT
projection polynomial fY

```

Why build this?

Suppose the input is :

$$f(x_1, x_2, x_3) = 5x_1 + 3x_2x_3 - 2x_1x_3$$

Now $X = \{1\}$

That means: Eliminate x_1 , but preserve the best value x_1 could contribute for every possible assignment of the remaining variables.

The remaining variables are:

$$Y = \{2,3\}$$

Mathematically, we're trying to compute

$$f^Y(x_2, x_3) = \max_{x_1 \in \{0,1\}} f(x_1, x_2, x_3).$$

Here we don't just want to evaluate that function for some inputs.

Here we want the program to actually return another polynomial involving only Y.

The output should eventually be:

$$f^Y(x_2, x_3) = 5 - 2x_3 + 3x_2x_3.$$

Why compute a projection polynomial at all?

Start with the optimization problem:

$$\max_{x \in \{0,1\}^V} f(x).$$

Suppose we divide the variables into

$$V = X \cup Y.$$

The projection

$$f^Y(x^Y) = \max_{x^X \in \{0,1\}^X} f(x^X, x^Y)$$

eliminates the X -variables without losing their optimal contribution.

That gives us a smaller optimization problem involving only Y:

$$\max_{x^Y} f^Y(x^Y).$$

So instead of continuing to optimize over every original variable, we optimize away a selected group X and represent their best possible contribution as a new function of the remaining variables Y.

So there's a computational challenge hiding inside that:

We don't merely want a table containing the value of f^Y for every Y-assignment. We want to construct f^Y explicitly as a polynomial.

Example to understand if this works

Polynomial $\rightarrow$ X,Y $\rightarrow$ P,Q,R $\rightarrow$ T $\rightarrow$ signature $\rightarrow \beta_y \rightarrow \mu(y)$
 $\rightarrow$ projection

$$5x_A + 3x_A x_B + 2x_C$$

$X = \{A\} \rightarrow \text{eliminate}$ $Y = \{B, C\}$

$$f^Y(x_B, x_C) = \max_{x_A \in \{0,1\}} f(x_A, x_B, x_C)$$

Step 1: Turn the polynomial into sets

Look at each monomial

$$5x_A \leftrightarrow \{A\}$$

$$3x_A x_B \leftrightarrow \{A, B\}$$

$$2x_C \leftrightarrow \{C\}$$

$\therefore$ Our hyperedges are:

$$\mathcal{H} = \{\{A\}, \{A, B\}, \{C\}\}$$

This is the monomial $\leftrightarrow$ set

Step 2: P, Q, R

Remember: P : completely inside X
 Q : crosses X & Y
 R : completely inside Y

Since

$$X = \{A\}, \quad Y = \{B, C\}$$

we get:-

$$\begin{aligned} \{A\} &\rightarrow P & \{A, B\} &\rightarrow Q \\ \{C\} &\rightarrow R \end{aligned}$$

Polynomial splits into:

$$\alpha = 500A$$

$$\beta = 300A \times B$$

$$\gamma = 200C$$

So:

$$f = \alpha + \beta + \gamma$$

Step 3: Construct T

This is where the signature starts.

$$Q = \{\{A, B\}\}$$

Intersect it with: $Y = \{B, C\}$

$$\therefore \{A, B\} \cap \{B, C\} = \{B\}$$

$$\text{So: } T = \{\{A, B\}\}$$

Notice c isn't in T
why? Because c wasn't involved in crossing term.

Step 4 - Signature

Since $T = \{B\}$

only one relevant T -monomial x_B

$\therefore$ Signature only has $\pm$ position.

If: $x_B = 0$
symbol given for signature
then

$$y = (0)$$

If: $x_B = 1$

then

$$y = (1)$$

y -assignment determines them.

Notice that x_c does not affect this signature.

For example:

$$(B, c) = (0, 0) \rightarrow y = (0)$$

$$\text{and } (B, c) = (0, 1) \rightarrow y = (0)$$

Two different full y -assignments have the same signature because, for the ~~case~~ crossing point, only B matters.

That's why we called the signature the relevant ON/OFF pattern.

Steps: why does this help with β ?

$$\beta = 3x_A x_B$$

Rewrite

$$\beta = 3x_A (x_B)$$

X-part is: x_A

Y-part is: x_B

But our signature already tells us whether x_B is 0 or 1.

Signature $y = (0)$

That tells us: $x_B = 0$

$$\therefore \beta_y = 3x_A(0) = 0 \quad \text{so } \underline{\beta_0 = 0}$$

Signature $y = (1)$

$$x_B = 1$$

$$\beta = 3x_A(1) = \underline{3x_A} \quad \text{so } \underline{\beta_1 = 3x_A}$$

~~That~~ This is why signature is helpful.

Once we know the signature, the relevant Y-part of the crossing term has become a known 0 or 1.

$\therefore \beta_y$ now depends only on x .

Step 6: Now compute $\mu(y)$

Remember: $x = x_A$

We define: $\mu(y) = \max_{x_A \in \{0,1\}} [x + \beta_y]$

this is where we optimize the x -variables.

For signature: $y = (0)$ $\beta_0 = 0$

$$\therefore \mu(0) = \max_{x_A} [5x_A + 0]$$

Try $x_A = 0$

0

$x_A = 1$

5

Maximum

$$\mu(0) = 5$$

Now signature

$y = (1)$

$$\beta_1 = 3x_A$$

$$\mu(1) = \max_{x_A} [5x_A + 3x_A]$$

$$= \max_{x_A} [8x_A]$$

Try $x_A = 0$

$x_A = 1$

8

Maximum

$$\mu(1) = 8$$

Signature does not store the optimized value.

Signature: (0) or (1) stores the ON/OFF pattern.

$$\text{Then } \mu(0) = 5 \quad \mu(1) = 8$$

stores the best x -contribution associated with these signatures.

Step: Don't forget γ

$$\gamma = 2x_C$$

why didn't we maximize that? because $C \in Y$.
we're keeping it.

So projection consist of

best contribution from x /

crossing part + r

We know: if $x_B = 0$, $\mu = 5$

if $x_B = 1$, $\mu = 8$

what polynomial in binary x_B gives those values?

Start with $g(x_B) = a + bx_B$

Since: $g(0) = 5$ we get $a = 5$

And $g(1) = 8 \leftarrow 5 + b = 8 \quad b = 3$

$\therefore g(x_B) = 5 + 3x_B$

Then add back: $r = 2x_C$

$$\boxed{5 + 3x_B + 2x_C}$$

Let's verify it directly.

Original: $f = 5x_A + 3x_A x_B + 2x_C$

Suppose:

$$B = 0 \quad C = 1$$

Original: $f = 5x_A + 0 + 2 =$

$$\text{Try } A=0: \quad 2$$

$$\text{Try } A=1 \quad 7$$

Best: 7

Our projected polynomial

$$f_y = 5 + 3x_B + 2x_C$$

$$B=0$$

$$, C=1$$

$$f_y = 5 + 3(0) + 2(1)$$

$$= 5 + 2 = 7$$

$$B=1$$

$$, C=1$$

$$f_y = 5 + 3(1) + 2(1)$$

$$= 10$$

So eliminating A didn't lose the optimal value.

So

$$f = 5x_A + 3x_Ax_B + 2x_C$$

↓

$$X = \{A\} \quad Y = \{B, C\}$$

↓

$$P = \{A\} \quad Q = \{A, B\} \quad R = \{C\}$$

↓

$$T = \{B\}$$

↓

signatures (0), (1)

$$B_0 = 0$$

$$B_1 = 3x_A$$

$$\begin{array}{c} \downarrow \\ \mu(0) = 5 \quad \mu(1) = 8 \\ \downarrow \end{array}$$

$$f^4 = 5 + 3x_B + 2x_C$$

That's projection using the signature idea.

Multiple T-set

$$f = 5x_A + 3x_A x_B + 4x_A x_C + 2x_D$$

with

$$X = \{A\} \quad \text{eliminating}$$

$$Y = \{B, C, D\}$$

Look only at the crossing terms;

$$3x_A x_B \text{ and } 4x_A x_C$$

Their hyperedge are

$$\{A, B\} \text{ and } \{A, C\}$$

Take their Y -parts:

$$\{A, B\} \cap Y = \{B\}$$

$$\{A, C\} \cap Y = \{C\}$$

$$\therefore \{\{B\}, \{C\}\} = T$$

Now we have 2 T-sets.
Means our signature has 2 positions.
 $y = (y_B, y_C)$

or more simply,

So possible signatures are:

$(0,0), (0,1), (1,0), (1,1)$

Now here's the important part:

Suppose: $y = (1,1)$

That means both T-monomials are active:

$$x_B = 1, \quad x_C = 1$$

When the projection formula constructs the polynomial corresponding to this combination, these active monomials can multiply:

Look at their sets $\{B\}$ and $\{C\}$

$$\text{Union is } \boxed{\{B\} \cup \{C\} = \{B, C\}}$$

Monomial $x_B x_C$

That's exactly the connection to your previous project.

active T-sets $\rightarrow$ take UNION $\rightarrow$ result tells us the T-monomial

For example:

$$(1,0) \rightarrow \{B\} \rightarrow x_B$$

$$(0,1) \rightarrow \{C\} \rightarrow x_C$$

$$(1,1) \rightarrow \{B\} \cup \{C\} = \{B,C\} \rightarrow x_B x_C$$

And: $(0,0) \rightarrow \emptyset \rightarrow 1$

the empty set corresponds to the constant monomial.

(This is where your union generator fits.)

$$(1,1) = p$$